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1. (a) Assume that for all $n \geq 1$

$$2 + 5 + 8 + \cdots + (3n - 1) = \frac{n(3n + 1)}{2}$$

First we prove that at $n = 1$ the left and right side are equivalent:
 $P(1) = P(1)$. When $n = 1$:

$$(3(1) - 1) = 2 = \frac{1(3(1) + 1)}{2} = \frac{4}{2} = 2$$

Assuming $P(k)$ is true. For $n \geq 1$, $P(k) \rightarrow P(k + 1)$ is again equivalent for the left and right side. RHS $P(k + 1)$

$$\frac{(k + 1)(3(k + 1) + 1)}{2} = \frac{(k + 1)(3k + 4)}{2} = \frac{3k^2 + 7k + 4}{2}$$

Since the LHS and RHS are equivalent we add the next element of the LHS to both sides and it should be equivalent to the RHS $P(k + 1)$

$$\begin{aligned} 2 + 5 + 8 + \cdots + (3k - 1) + (3(k + 1) - 1) &= \frac{k(3k + 1)}{2} + (3(k + 1) - 1) \\ &= \frac{(k)(3k + 1)}{2} + \frac{6k + 4}{2} \\ &= \frac{3k^2 + k}{2} + \frac{6k + 4}{2} \\ &= \frac{3k^2 + 7k + 4}{2} \end{aligned}$$

The last equation shows that $P(k + 1)$ is true under the assumption that $P(k)$ is also true. Therefore,

$$2 + 5 + 8 + \cdots + (3n - 1) = \frac{n(3n + 1)}{2}$$

Assume that for all $n \geq 1$

- (b) Assume for all $n \geq 1$:

$$\prod_{k=1}^n \left(1 + \frac{1}{k}\right) = n + 1$$

First we prove $P(k)$ is true such that at $n = 1$ the left and right side are equivalent:

$$\prod_{k=1}^1 \left(1 + \frac{1}{k}\right) = 1 + 1$$

$$(1 + 1) = (1 + 1)$$

$$2 = 2$$

Since $P(k)$ is true we will now prove that $P(k) \rightarrow P(k + 1)$ is again equivalent for the left and right side. LHS:

$$= (k + 1) + 1 = k + 2$$

RHS and LHS:

$$\prod_{k=1}^n \left(1 + \frac{1}{k+1}\right) = (k+1)\left(1 + \frac{1}{k+1}\right)$$

$$= k + 1 + \frac{k+1}{k+1} = k + 1 + 1 = k + 2$$

This proves that $P(k+1)$ is true assuming that $P(k)$ is true. Therefore, by mathematical induction,

$$\prod_{k=1}^n \left(1 + \frac{1}{k}\right) = n + 1$$

for all $n \geq 1$.

2. A license plate is a string over the alphabet of 36 characters consisting of 26 uppercase Latin letters and 10 numerals (0 through 9). (2 points for a correct expression and 2 points for justification)

- (a) How many 7-character license plates have at least 2 numerals?

The total number of combinations for 7 positions and 36 characters is 36^7 . To find the case where there is at least 2 we can find the compliment (number of combos with no numbers or one number) and subtract that from the total.

- Total number with no numbers: 26^7
- Total number with one number: $\binom{7}{1} \cdot 10 \cdot 26^6$

Number of license plates that have at least 2 numerals:

$$= 36^7 - 26^7 - \binom{7}{1} \cdot 10 \cdot 26^6$$

- (b) How many 7-character license plates contain the word MILES as a consecutive substring?

To find the possible combos, MILES is a 5 character string that must be in order in the license plate thus there are only 3 ways to place the specific string on the license plate. There are always 2 empty characters that allows for 36 characters each thus:

- Number of MILES placements: 3
- Number of unique options for 2 spots with 36 characters: 36^2

Number of plates that contain the word MILES as a consecutive substring:

$$= 3 \cdot 36^2$$

- (c) How many 7-character license plates consist of 3 different characters with 3 copies of one, 3 copies of another and a single copy of the last? Ex: AAA3331, P33T3TT, ...

Choose 3 characters. Two of them appear 3 times, and the other appears once.

- Choose 3 characters from 36: $\binom{36}{3}$
- Choose which of those 3 appears once: 3
- Arrange 7 spots with those repetitions: $\frac{7!}{3! \cdot 3! \cdot 1!}$

Number of ways to arrange 3 different characters with 3 copies of one, 3 copies of another and a single copy of the last:

$$= \binom{36}{3} \cdot 3 \cdot \frac{7!}{3! \cdot 3! \cdot 1!}$$

- (d) How many 7-character license plates have exactly 3 different numerals and 4 copies of the same letter?

Similar to the previous one you are choosing 1 character but in this case an alphabetical character from 26 which is repeated 4 times also in this problem you have the last 3 characters as unique numerals

- Choose the letter from 26: $\binom{26}{1}$
- Choose and order 3 different numerals: $P(10, 3)$
- Arrange 7 spots with that letter repeated 4 times: $\frac{7!}{4! \cdot 1!}$

The number of ways to have exactly 3 different numerals and 4 copies of the same letter:

$$= \binom{26}{1} \cdot P(10, 3) \cdot \frac{7!}{4! \cdot 1!}$$

- (e) How many 7-character license plates are there that are all letters and they are in alphabetical order?

For this one you choose 7 alphabetical characters from 26 and that set has one alphabetical order.

- Choose the 7 from 26: $\binom{26}{7}$

The number of ways that the license plate is all letters and they are in alphabetical order:

$$= \binom{26}{7}$$